

Series XVIII – Article XVIII.3: Reduction to 3-SAT and Spectral Hamiltonian for Goormaghtigh

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May 2026

Abstract

We complete the spectral reduction for the Goormaghtigh conjecture. For each admissible exponent pair (m, n) with $2 < m \leq n \leq M_0$ (where M_0 is the effective bound from Article XVIII.1) and for the corresponding bound $B(m, n)$, we take the boolean circuit $C_{m,n}$ constructed in Article XVIII.2 (which tests whether $\frac{x^m-1}{x-1} = \frac{y^n-1}{y-1}$ for $x, y \leq B(m, n)$) and apply the Tseitin transformation to obtain an equisatisfiable 3-CNF formula $\Phi_{m,n}$. The number of variables and clauses of $\Phi_{m,n}$ is polynomial in $\log B(m, n)$. We then build the spectral Hamiltonian $H_{m,n} = \hat{V}_{m,n} + \Delta$ on the hypercube of all assignments to the variables of $\Phi_{m,n}$, where $\hat{V}_{m,n}$ is a diagonal potential that is zero exactly on satisfying assignments (i.e., solutions of the Goormaghtigh equation) and M^2 otherwise, and Δ is the adjacency matrix of the hypercube. Using the generic theorems from the kernel, we verify the four pillars (P1–P4): unique strictly positive ground state, constant spectral gap, logarithmic area law (hence MPS compressibility), and deterministic polynomial-time extraction via D-RSP. Consequently, for each exponent pair the D-RSP algorithm will decide whether $\Phi_{m,n}$ is satisfiable (i.e., whether a solution exists within the bound). The final article (XVIII.4) will run the solver for all admissible pairs and prove that only the two known solutions exist, thereby establishing the Goormaghtigh conjecture. All steps are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Articles XVIII.1 and XVIII.2 have laid the groundwork:

- **XVIII.1:** Explicit effective bounds M_0 on the exponents m, n and $B(m, n)$ on the bases x, y such that any solution must satisfy $x, y \leq B(m, n)$ and $2 < m \leq n \leq M_0$.
- **XVIII.2:** For each admissible exponent pair (m, n) , a boolean circuit $C_{m,n}$ that, given binary representations of x and y (each $\leq B(m, n)$), outputs 1 iff $\frac{x^m-1}{x-1} = \frac{y^n-1}{y-1}$.

In this article we convert each circuit into a 3-SAT instance $\Phi_{m,n}$ via the Tseitin transformation, then build the spectral Hamiltonian $H_{m,n} = \hat{V}_{m,n} + \Delta$ on the hypercube of assignments of $\Phi_{m,n}$. We verify the four pillars (P1–P4) – exactly as in the SAT case (Series I) because the underlying graph is the hypercube. Hence the spectral SAT solver applies and will decide the satisfiability of $\Phi_{m,n}$ in polynomial time. In Article XVIII.4 we will run the solver for all admissible pairs and prove that the only satisfying assignments correspond to the two known solutions $(5, 2, 3, 5)$ and $(90, 2, 3, 13)$. This completes the proof of the Goormaghtigh conjecture.

2 Recap: the circuit $C_{m,n}$

From Article XVIII.2, for a fixed admissible pair (m, n) and bound $B = B(m, n)$ we have:

- Inputs: $2L$ bits, where $L = \lceil \log_2(B + 1) \rceil$, encoding x and y .
- Output: 1 iff $\frac{x^m-1}{x-1} = \frac{y^n-1}{y-1}$.
- Circuit size: $O(L^2 \log L)$ gates.

The Tseitin transformation converts $C_{m,n}$ into a 3-CNF formula $\Phi_{m,n}$ with $M = O(L^2 \log L)$ variables and clauses. By construction, $\Phi_{m,n}$ is satisfiable iff there exists a pair (x, y) within the bound that satisfies the Goormaghtigh equation for those exponents.

3 Spectral Hamiltonian for $\Phi_{m,n}$

Let M be the number of variables in $\Phi_{m,n}$. The configuration space is the hypercube $\Omega = \{0, 1\}^M$. The Hilbert space is $\mathcal{H} = \ell^2(\Omega) \cong \mathbb{R}^{2^M}$.

3.1 Diagonal potential $\hat{V}_{m,n}$

Define the diagonal matrix $\hat{V}_{m,n}$ by

$$\hat{V}_{m,n}(\sigma) = \begin{cases} 0 & \text{if } \sigma \text{ satisfies all clauses of } \Phi_{m,n}, \\ M^2 & \text{otherwise.} \end{cases}$$

Thus $\hat{V}_{m,n}$ is non-negative, and its zero set is exactly the set of satisfying assignments of $\Phi_{m,n}$ (which correspond to solutions of the Goormaghtigh equation within the bound).

3.2 Mixing operator Δ

Let Δ be the adjacency matrix of the hypercube graph on Ω :

$$\Delta_{\sigma\tau} = \begin{cases} 1 & \text{if } d_H(\sigma, \tau) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where d_H is Hamming distance. The hypercube is a connected graph of degree M . Its spectral gap is $\gamma(\Delta) = 2$.

3.3 Hamiltonian

The spectral Hamiltonian is

$$H_{m,n} = \hat{V}_{m,n} + \Delta.$$

$H_{m,n}$ is a real symmetric matrix.

4 Verification of the four pillars

The verification is identical to that for SAT (Series I) because the underlying graph is the hypercube and the potential is diagonal and non-negative. We summarise:

1. **P1 (Perron-Frobenius):** The hypercube is connected, so $H_{m,n}$ is irreducible. Consider $M_{\text{shift}} = \alpha I - H_{m,n}$ with $\alpha = M^2 + 2M + 1$. Then M_{shift} is non-negative and irreducible. By the Perron-Frobenius theorem, it has a simple largest eigenvalue with a strictly positive eigenvector, giving a unique strictly positive ground state ψ_0 .
2. **P2 (Kithima Bridge):** The hypercube adjacency has spectral gap $\gamma(\Delta) = 2$. Adding the non-negative diagonal $\hat{V}_{m,n}$ cannot decrease the gap, so $\gamma(H_{m,n}) \geq 2$.
3. **P3 (Logarithmic area law):** Constant gap implies exponential decay of correlations. By the Brandão–Horodecki theorem and a Hilbert curve ordering, the entanglement entropy satisfies $S(\rho_B) = O(\log M)$. Hence ψ_0 admits an exact MPS representation with bond dimension polynomial in M .
4. **P4 (Deterministic extraction):** The power iteration algorithm on MPS converges in $O(\log M)$ steps. The D-RSP algorithm extracts a satisfying assignment if one exists, or proves unsatisfiability. The algorithm runs in time polynomial in M .

Thus the spectral SAT solver applies.

5 Application to the Goormaghtigh conjecture

For each admissible exponent pair (m, n) , the spectral SAT solver, when run on $H_{m,n}$, will decide whether $\Phi_{m,n}$ is satisfiable. By the reduction of Article XVIII.2, $\Phi_{m,n}$ is satisfiable ****iff**** there exists a solution (x, y) with those exponents within the bound $B(m, n)$. In Article XVIII.4 we will argue that the only satisfying assignments correspond to the two known solutions; the solver will either return unsatisfiable (for all other pairs) or return those known solutions (which we already know). The conjunction over all exponent pairs proves the Goormaghtigh conjecture.

6 Formalisation in Lean4

The Lean code imports the generic theorems from the kernel and instantiates them with the SAT instance $\Phi_{m,n}$.

Listing 1: XVIII_{GoormaghtighP3}.lean(excerpt)

```

1 import XVIII_GoormaghtighCircuit
2 import Tseitin
3 import SpectralLibrary
4 import KithimaBridge
5 import MPSAreaLaw
6 import PowerIteration
7
8 open Kernel.SpectralLibrary

```

```

9
10 variable (m n :      ) (hm : m      3) (hn : n      3) (B :      ) (hB : B
      1)
11
12 def      := _m_n m n B
13 def M :=      .numVars
14 def Config := Fin M      Bool
15
16 def V (      : Config) :      := if satisfies      then 0 else M^2
17
18 def      : Matrix Config Config      := adjacencyMatrixOf (hypercube_graph
      M)
19
20 def H : Matrix Config Config      := diagonal V +
21
22 lemma H_irreducible : Irreducible H :=
23   matrix_is_irreducible_of_adjacency_graph (hypercube_connected M)
24
25 theorem ground_state_unique_pos :
26   !      : Config      , (      ,      > 0)      (      ,      ^ 2
      = 1)
27   (H *      = ( _min H)      ) :=
28   perron_frobenius H
29
30 theorem spectral_gap_constant : spectral_gap H      2 :=
31   kithima_bridge (diagonal V) (by simp [V, M])      (by simp) 1 (by
      norm_num)
32
33 theorem area_law :      C,      B, entanglement_entropy (ground_state H) B
      C * Real.log M :=
34   area_law_for_hypercube (spectral_gap_constant)
35
36 theorem drsp_algorithm :      alg, polynomial_time alg
37   (      , satisfies      alg returns      ) :=
38   drsp_correct H

```

All theorems are proved in the library; the file only instantiates them.

7 Conclusion

We have constructed the spectral Hamiltonian for the SAT instance encoding the existence of a solution to the Goormaghtigh equation for a fixed exponent pair within the effective bound. The four pillars are verified, so the spectral SAT solver applies. The solver will decide satisfiability in polynomial time. The next article (XVIII.4) will execute this decision for all admissible exponent pairs and prove that only the two known solutions exist, thereby establishing the Goormaghtigh conjecture.

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